ReguMind — Requirements Gathering

1) Stakeholder Analysis

Stakeholder	Needs	Expectations from System
Students	Quick, clear answers about regulations	Accurate answers with citations in simple language
University Administration	Reliable representation of official regulations	Data isolation, correctness, easy document upload
Faculty Members	Easy access to bylaws and policies	Searchable regulations and precise retrieval
System Developers	Maintainable and scalable architecture	Modular design, clear APIs, reusable components
Project Supervisor	Proper application of AI concepts	Working RAG system with evaluation metrics

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2) User Stories & Use Cases

User Stories

- As a student, I want to ask questions about academic regulations and receive accurate answers with references.
- As an admin, I want to upload regulation PDFs so the system can index them automatically.
- As a faculty member, I want to search regulations semantically instead of reading long documents.
- As a developer, I want a modular pipeline so I can improve retrieval or generation independently.

Use Cases

Use Case	Actor	Description
Ask regulation question	Student	User enters a question and receives a cited answer
Upload regulation document	Admin	PDF is processed, chunked, embedded, and indexed
Retrieve relevant chunks	System	Hybrid retrieval finds best matching sections
Generate grounded answer	System	LLM generates response strictly from retrieved chunks
View citations	Student	User can see the exact regulation source

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3) Functional Requirements

- Upload and process regulation PDF documents.
- Intelligent document chunking and preprocessing.
- Generate semantic embeddings for document chunks.
- Store embeddings in a vector database (FAISS/Chroma).
- Implement Hybrid Retrieval (Dense + BM25).
- Apply re-ranking on retrieved chunks.
- Generate answers using RAG with citation support.
- Provide a web interface for users to ask questions.
- FastAPI backend to manage AI pipeline and requests.
- Multi-tenant architecture to isolate university data.

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4) Non-Functional Requirements

Performance

- Response time should be less than 2 seconds per query.
- Efficient indexing for large regulation documents.

Security

- Complete isolation between university datasets (multi-tenant).
- Secure handling of uploaded documents.

Usability

- Simple web interface for asking questions and viewing answers.
- Clear display of citations and references.

Reliability

- System should return grounded answers with minimal hallucination.
- Stable API performance during repeated queries.
- Proper logging for debugging and evaluation.